

1. คำนวณค่ากระแส $i(t)$

พิจารณาที่โนด จะพบว่า $i(t) = i_s(t) + \frac{v(t)}{R_L}$

ที่เวลา $t = 0$ s ; $i(0) = i_s(0) + \frac{v(0)}{R_L} = 0.322589559150028 + \frac{4.17410440849972}{10} = 0.74$ A ✓

ที่เวลา $t = 10$ s ; $i(10) = i_s(10) + \frac{v(10)}{R_L} = 0.32379514292850 + \frac{4.16204857071497}{10} = 0.74$ A ✓

ที่เวลา $t = 100$ s ; $i(100) = i_s(100) + \frac{v(100)}{R_L} = 0.33191477675941 + \frac{4.08085223240593}{10} = 0.74$ A ✓

จะเห็นว่า ค่าของกระแส $i(t)$ มีค่าคงที่เท่ากับ 0.74 A ตลอดทุกช่วงเวลา

$i = \frac{dq}{dt} \rightarrow q = \int i dt$

2. ค่าประจุไฟฟ้า $q(t)$ และ Q_n

จาก $q(t) = \int_0^t i(\alpha) d\alpha = \int_0^t 0.74 d\alpha = 0.74(\alpha) \Big|_0^t = 0.74t$ [C]

และ $Q_n = \int_0^{t_n} i(\alpha) d\alpha = \int_0^{3600} 0.74 d\alpha = 2664$ [C]

$0.74 \int_0^{3600} 1 d\alpha = 0.74(\alpha) \Big|_0^{3600} = 0.74(3600) = 2664$ C ✓

3. จากสมการ $v_s(t) = E_0 - K \cdot q(t) + A_a e^{-B_a q(t)} - A_b e^{-B_b(2664 - q(t))}$

และ KVL วงปิดซ้าย ; $v(t) = v_s(t) - i(t)R_i = \left[E_0 - K \cdot q(t) + A_a e^{-B_a q(t)} - A_b e^{-B_b(2664 - q(t))} \right] - 0.74 R_i$

$v(t) = (E_0 - 0.74 R_i) - K \cdot q(t) + A_a e^{-B_a q(t)} - A_b e^{-B_b(2664 - q(t))}$

$v_s(t) = E_0 - K \int_0^t i(\alpha) d\alpha + A_a e^{-B_a \int_0^t i(\alpha) d\alpha} - A_b e^{-B_b \int_t^{t_n} i(\alpha) d\alpha}$

$Q_n = \int_0^{t_n} i dt = \int_0^t i dt + \int_t^{t_n} i dt$
 $Q_n = q(t) + (Q_n - q(t))$

$v(t) = E_0 - K q(t) + A_a e^{-B_a q(t)} - A_b e^{-B_b(2664 - q(t))}$

$-V_s(t) + i(t)R_i + v(t) = 0$

$v(t) = V_s(t) - R_i i(t)$

$v(t) = V_s(t) - 0.74 R_i$

$$V(t) = E_0 - k q(t) + A_a e^{-\beta_a q(t)} - A_b e^{-\beta_b (2664 - q(t))} - 0.74 R_i$$

(4)

$$V(t) = (E_0 - 0.74 R_i) - k q(t) + A_a e^{-\beta_a q(t)} - A_b e^{-\beta_b (2664 - q(t))}$$

$$y = mx + c$$

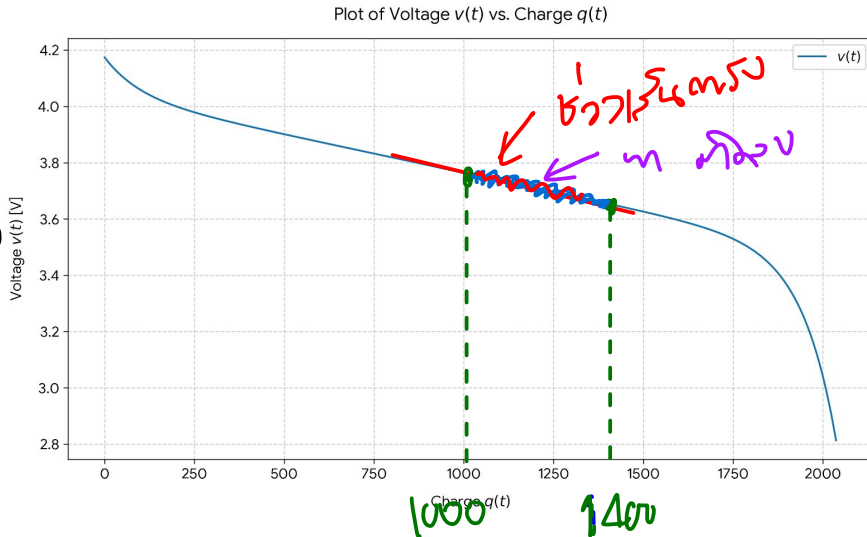
$$V(t) = m q(t) + c$$

$q(t)$ slope

$$e^{-\beta_a q(t)} \approx 0$$

$$e^{-\beta_b (2664 - q(t))} \approx 0$$

$q(t)$



$$V(t) = (E_0 - 0.74 R_i) - k q(t)$$

$$V(t) = E'_0 - k q(t)$$

$$\therefore V(t) = -k q(t) + E'_0$$

$q = 0.74 t$

$$k = \frac{\Delta V}{\Delta q} \text{ (V/C)}$$

$$= \frac{V(1000) - V(1400)}{q(1000) - q(1400)}$$

$$-k = -2.724 \times 10^{-4} \text{ V/C}$$

$$k = 2.724 \times 10^{-4} \text{ V/C}$$

$$\therefore V(t) = -2.724 \times 10^{-4} q(t) + E'_0$$

$$V(1000) = -2.724 \times 10^{-4} q(1000) + E'_0$$

$$3.8353 = -2.724 \times 10^{-4} (740) + E'_0 \Rightarrow E'_0 = 4.036876 \text{ V}$$

$$E'_0 = E_0 - 0.74 R_i \quad (2)$$

$$R_i = \frac{E_0 - E'_0}{0.74} = \frac{4.0801 - 4.036876}{0.74} = 0.0584 \, \Omega$$

$$V(100) \quad \#$$

$$Q_n - q(t) = \int_0^t \dots \int_0^t \dots \int_0^t \dots$$

$$\lim_{t \rightarrow 0} A_a \left(\frac{-\beta_a q(t)}{e^{\beta_a q(t)}} = 1 \right)$$

$$t=0; q(t)=0$$

$$t=0 + (w, q(t) \text{ w})$$

$$e^{-\beta_a q(t)} \text{ due to } 2000$$

$$V(t) = (E_0 - 0.74 R_i) - K q(t) + A_a e^{-\beta_a q(t)} - A_b e^{-\beta_b (2664 - q(t))}$$

$$V(0) = E_0 - 0.74 R_i + A_a$$

$$4.1741 = 4.03687 + A_a \rightarrow A_a = 0.13723 \, V$$

$$V(t) = E'_0 - K q(t) + A_a e^{-\beta_a q(t)} - A_b e^{-\beta_b (2664 - q(t))}$$

$$t = 100 \, s; V(100) = 4.03687 + 0.13723 e^{-\beta_a (74)} - 0$$

$$\beta_a = 0.01533 \, 1/C$$

$$V(t) = E'_0 - K q(t) + A_a e^{-\beta_a q(t)} - A_b e^{-\beta_b (2664 - q(t))}$$

$-0.01533 \times 2700 \times 0.24 = 0$

min A_b, β_b for min $V(t) \approx 0$, min $S_1 q$ ($t > 2700$ s)

$t = 2700$; $q(2700) = 2008$

$$\rightarrow V(2700) = E'_0 - K q(2700) + 0 - A_b e^{-\beta_b (666)}$$

$$3.05 = 4.03687 - 0.5449 - A_b e^{-666 \beta_b}$$

$$A_b e^{-666 \beta_b} = 0.44257 \quad \text{--- (3) } \checkmark$$

$t = 2750$; $q(2750) = 2035$

$$V(2750) = 4.03687 - 0.5549 + 0 - A_b e^{-\beta_b (629)}$$

$$2.8244 = 4.03687 - 0.5549 + 0 - A_b e^{-\beta_b (629)}$$

$$A_b e^{-\beta_b (629)} = 0.6582 \quad \text{--- (4)}$$

(3) / (4) :

$$\frac{A_b e^{-666 \beta_b}}{A_b e^{-629 \beta_b}} = \frac{0.44257}{0.6582}$$

$$\beta_b = 0.0107 \text{ 1/C}$$

$$A_b = 551.776 \text{ V}$$

$$A_{v_e} = \frac{-\beta (2664 - 987)}{551.176 \text{ e}} - 0.0167 (2664 - 2035) = 0.65V$$